MCQ A · scalar product and orthogonality

Orthonormal coordinate system. After S51; MCQ B after S56.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

1. $(1;2;-1)\cdot(2;0;3)$ is:

A. 5 B. -1

C. 1 D. -3

2. The norm of $(2;-1;2)$ is:

A. 9 B. 5

C. $\sqrt{5}$ D. 3

3. To prove that a vector is normal to a plane, it is enough to test that it is orthogo

A. two non-collinear directions of the plane B. a direction of the plane

C. two collinear directions of the plane D. a normal vector to the plane

4. Two nonzero vectors with a negative scalar product form an angle that is:

A. acute B. right

C. obtuse D. zero

5. Two orthogonal lines in space:

A. always intersect B. may be non-coplanar

C. are parallel D. coincide

MCQ B · projection, minimum and volume

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. The projection of $M(2;3;4)$ onto the plane $z=0$ is:

A. $(0;0;4)$ B. $(2;3;4)$

C. $(0;0;0)$ D. $(2;3;0)$

7. H is the projection of M onto a plane. For every point P of the plane:

A. $MP^2 = MH^2 + HP^2$ B. $MP = MH + HP$

C. $MH^2 = MP^2 + HP^2$ D. $MP = HP$

8. A tetrahedron with base area 6 and height 4 has volume:

A. 24 B. 12

C. 8 D. 6

9. For vectors u, v unit vectors such that $u \cdot v = -\frac{1}{2}$, the angle is:

A. $\pi/3$ B. $2\pi/3$

C. $\pi/2$ D. π

10. $A(1;0;0)$, $M(0;2;3)$, and d with direction $(1;1;0)$ through A. The projection of M on d is:

A. $(1;1;0)$ B. $(0;2;0)$

C. $(1;0;0)$ D. $(\frac{3}{2};\frac{1}{2};0)$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every point of a plane is closer to M than its projection.

2. Two vectors with zero scalar product are collinear.

3. Calculating a volume from two different faces must give the same result.